

DUAL-INPUT NON-ISOLATED DC-DC CONVERTER WITH VEHICLE TO GRID FEATURE

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Abstract— Electric Vehicles (EVs) become a more prominent solution to replace all fossil fuel-based combustion vehicles. During the parking of EVs, power produced by solar photovoltaic (PV) present in the PV powered EVs is underutilized when the capacity of the EV battery is full. Also, a converter is devoted in the conventional EVs to perform the vehicle to grid (V2G) or vehicle to vehicle (V2V) operation. To utilize PV and to perform V2G operation, a novel non-isolated dual-input single output DC-DC converter (DISOC) is proposed. The DISOC structure can be reconfigured to perform six types of operation based on the status of power availability with PV, battery and also the running status of the EV. Simultaneous power transfer from both the input sources, charging the battery from solar PV, V2G and G2V operations are the key features of the proposed converter. The converter operation, component design, effect of parasitic elements on the converter performance, small-signal model, etc., have been reported. The hardware prototype of the converter is fabricated for 500 W, and the experimental results are presented.

Keywords: *bidirectional converter; dual input-single output; electric vehicle; hybrid system; power converter.*

I. INTRODUCTION

Throughout the world, the focus is given for developing suitable technologies to provide clean energy to protect the environment. The sources such as solar PV, ultracapacitor, battery, etc., are utilized as an alternative to the conventional fossil fuel in the vehicles like car, bus, truck, ship, etc. A 'Hybrid Energy System (HES)' by integrating more than one energy source is highly preferred to provide reliable power in 'Electric Vehicles (EVs)'. A power electronic converter plays a major role in the integration of HES, and it delivers power from the input sources into

loads of the EV. Conventionally, parallel-connected single input-single output DC-DC converters are used in HES. The dedicated reactive elements, semiconductor switches and controller of each parallel-connected systems lead to increase the cost, size and complexity. To overcome these demerits, the 'Multi- Input single output DC-DC Converters (MICs)' is proposed.

The MICs are mainly classified into isolated and non isolated topologies. The isolated topologies use a high frequency transformer for the isolation purpose, and the number of input sources can be interfaced using multiple windings in the same core of the transformer. The isolated topologies have two or more converter stages between the input sources and the load. These converters require several semiconductor switches. Thus, these switches lead to high cost, more losses and massive in size. From the aforementioned drawbacks, the isolated MICs loss their attraction in some of the applications.

Various non-isolated MICs are reported in the past one and half decades. The reported MICs are intended to operate in buck, buck-boost and boost. A typical improved MIC buck topology is reported to transfer power simultaneously/independently from the input sources into the load. Most of the reported buck-boost converters can be operated in all three operations, i.e., buck/boost/buck boost using the same structure with appropriate control of their switches. The topologies of buck-boost MIC are reported, and these structures do not allow the simultaneous power transfer from the input sources. But, the structures proposed in and allow the simultaneous power transfer. A topology of MIC to operate in boost/buck boost is reported. But, two inductors and two capacitors are used to transfer the energy from the sources to the load.

Though the buck-boost converters have the capacity to operate in boost mode, a wide range of output voltage cannot be achieved in some topologies. Even, few

topologies require more switches to perform the boost operation. A dedicated boost MICs are reported. The MIC boost topology is derived by combining two single input boost converters in. A newly designed MIC topology with zero voltage switching is reported in. Here, two inductors are used to integrate two input sources. A boost MIC converter based on a switched-diode-capacitor voltage accumulator is presented. There are five capacitors and two inductors employed to combine the two input sources. A boost MIC for EV application is proposed. The second source alone cannot charge the inductor, and the voltage stresses of the switches are also more due to the negative polarity of the second input source. A new boost MIC is reported in, and it has a limited voltage gain. Similar MICs with two input sources are reported. These MICs utilize more components, and some of those MICs could perform the simultaneous power transfer from the connected input source.

The dual-input DC-DC converters with a high DC voltage gain are reported. Though the high DC voltage gain is achieved, those converters utilize more reactive components. Further, power transfer among the input sources, V2G, V2V, etc., are not possible in certain converters. To overcome certain aforementioned drawbacks such as utilization of more passive and active components and dedicated converters to perform operations like V2G and V2V, and achieving the simultaneous power transfer from the connected input sources, a novel DC-DC converter is proposed in this paper.

II. DISOC TOPOLOGY

Fig. 1 shows the proposed DISOC circuit for EV application, and the circuit contains three IGBTs with antiparallel diode (S_2 , S_m and S_{Bi}), one IGBT without antiparallel diode (S_1), a diode (D_1), an inductor (L), an output capacitor (C) and four relays (R_1 , R_2 , R_3 and R_4). The switch, i.e., IGBT without antiparallel diode S_1 is used to ensure unidirectional current flow. Otherwise, the source V_1 will be bypassed through the relay R_4 when the switch S_2 is turned on.

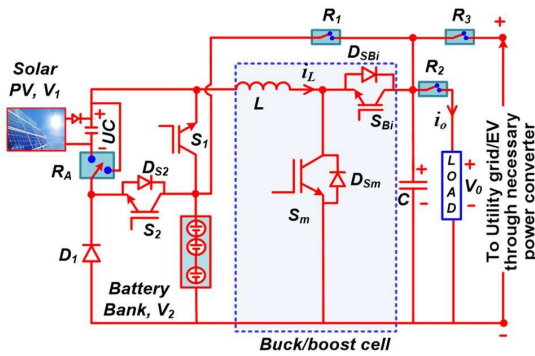


Fig. 1. The converter topology of DISOC in solar-powered vehicle

The main objective of this research work is to develop a novel dc-dc converter DISOC to integrate two input energy sources. The key features of the DISOC are highlighted as follows.

- i) The front-end converter structure is proposed using two semiconductor switches, a diode, and two relays. The structure empowers the proposed converter even when any one of the input source is not available. The switch S_2 enables the cascaded operation of the input sources, and achieves a higher output voltage.
- ii) Arrangement of relays with the power converter structure is unique in DISOC, and the arrangement of relays help to perform six modes of operation in the proposed DISOC.
- iii) The switch empowers the DISOC to operate the input sources in cascaded manner and helps to achieve more output voltage.
- iv) The relay R_3 enables the DISOC to perform the V2G and V2V program
- v) The relay R_1 enables the DISOC to charge V_2 from V_1 .
- vi) The position of the diode and the arrangement of the input source V_1 helps to avoid the floating of the input V_1 .
- vii) The proposed DISOC uses lower components and performs six modes of operation using the same converter structure.
- viii) The loss breakdown analysis and efficiency profile highlights that the proposed non-isolated dual-input single output DC-DC converter has lower power losses and higher efficiency compared to few conventional converters.

III. MODES OF OPERATION

Type i - C-boost operation in CCM: In this state, PV transfers the power individually to the load. In this The C-boost is one of the features of the proposed converter. Here, the relays R_2 and R_4 are in the on condition, and the other relays remain in the off state. The C-boost operation can be performed in three ways based on the input source which involves in the fourth mode, i.e., freewheeling operation of the inductor L . The three ways of operations are as follows:

- i) when V_1 freewheels
- ii) when V_1+V_2 freewheels
- iii) when V_2 freewheels.

The voltage waveform of the inductor L during three ways of operations are shown in Fig. Though the charging sequence of the three input sources changes in the first three modes of operations, the output voltage produced by the DISOC during C-boost is not affected. But, the involvement of the source in the fourth mode decides the output voltage. The analysis of the DISOC in C-boost is carried out in the CCM.

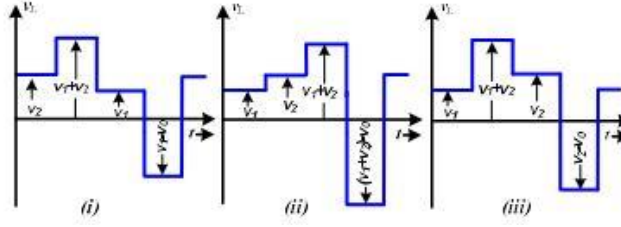


Fig.2. Fig. 2 Analytical waveform of inductor related to voltage involve during freewheeling operation (i) V_1 (ii) $V_1 + V_2$ (iii) V_2

The value of the inductor current is

$$I_L = I_{L0} + \frac{1}{L} V_2 d_1 T$$

Where $V_L = V_2$

Type i - C-boost operation in DCM: During the mode 4 operation, the energy of the inductor exhausted at $dxTs$, hence the amount of energy present in the inductor become zero. So, it results a DCM operation in the DISOC. Using the expression of inductor current during all four modes of operations, the value of dx is obtained. To simplify the calculation, the values of the voltages are considered as, $V_1 = aV_2$; $V_2 = V$.

$$d_x = \frac{V}{V_C - V} [ad_1 + (1+a)d_2 + d_3]$$

Using the average value of the capacitor current, the expression of DCM gain G_{DCM} is computed as

$$G_{DCM} = 0.5 + \sqrt{0.25 + \frac{[ad_1 + (1+a)d_2 + d_3]^2}{2\tau_L}}$$

where $\tau_L = Lfs/R$. τ_L is the normalized inductor time constant, and R is the load resistance. By equating the CCM and DCM gains, the boundary normalized inductor time constant τ_{LB} is obtained as given as

$$\tau_{LB} = \frac{D^2 K^2}{2M(M-1)}$$

$$\text{where } M = \frac{D[a(x+y) - x] + 1}{1-D}; K = ax + (1+a)y + 1$$

Type ii - Solar PV-powered operation: For type ii operation, input source V_1 alone delivers power into the load. During this type of operation, the relay R_2 and R_A are in the active state, and the other relays R_1 and R_3 are kept in the off state. Here, two modes of operations such as mode 5 and mode 6 present. In mode- 5 operation, the switch S_m is turned on and the inductor stores energy from the source V_1 . In mode-6, the switch S_m is turned off, and the stored energy in the inductor is freewheeled into the load with a voltage of $V_0 - V_1$. The equivalent circuits of each mode are shown in Fig. The value of $d_1 = d_2 = 0$ and $d_3 = D$. The duty ratio of S_m controls the output voltage of the DISOC. The output

voltage of the DISOC during type ii operation is expressed as

$$V_0 = \frac{V_1}{1-D}$$

Type iii - Battery-powered operation: For type iii operation, the input source V_2 alone delivers power into the load. Here also, two modes of operations such as mode 7 and mode 8 present. The relay R_2 is kept permanently in the on state, and the remaining relays are kept in the off state of the DISOC during type iii operation is expressed as,

$$V_0 = \frac{V_2}{1-D}$$

Type iv - Battery charging from solar PV in parking mode: To charge the battery from solar PV, the relays R_1 and R_A are kept on, and the rest of relays are kept in the off position. Here, when the solar PV charges the battery, two modes of operations such as mode 9 and mode 10 present in the DISOC. The output voltage of the DISOC during type iv operation when power delivered from solar PV to the battery, is expressed as,

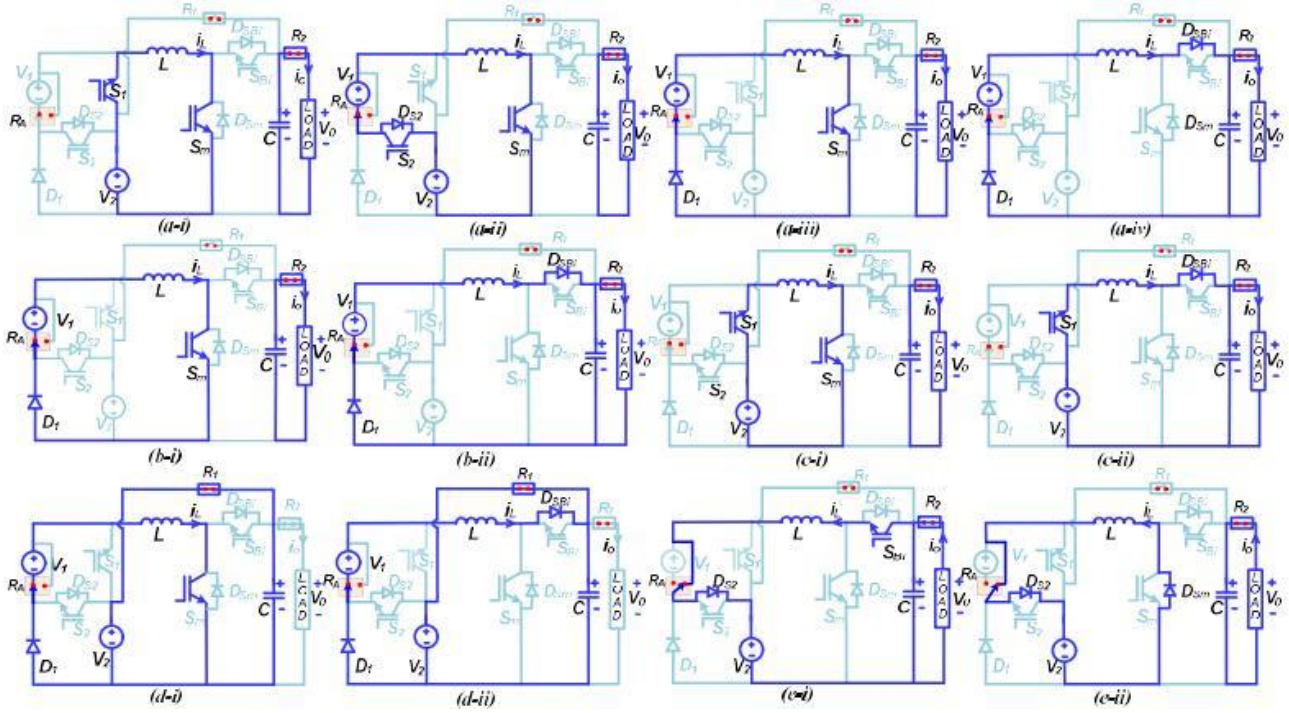
$$V_2 = \frac{V_1}{1-D}$$

During this operation, the output voltage of the DISOC is regulated to the battery charging voltage.

Type v - Power flow from V2V or V2G in parking mode: To export power from solar PV into the utility grid, the relay R_3 and R_A are kept in the on position, and rest of all the relays are kept in the off position. This Type v has two modes of operations such as mode 11 and mode 12. Here, the equivalent circuits are similar to Type II operation except the relay position. The duty ratio d_3 of the switch S_m controls the output voltage, which is required across the DC-link of the inverter. In practical case, the power generated from the solar PV is fed into the utility grid of three-phase/single-phase through the DISOC – three phase/single-phase inverter – appropriate filter circuit.. The output voltage of the DISOC in Type V operation is expressed as,

$$V_{Grid} = \frac{V_1}{(1-D)}$$

During this mode of operation, the output voltage of the DISOC is regulated to the DC-link voltage of the grid connected inverter. When the power is delivered from solar PV to the other EV to perform V2V mode of operation, the output voltage of the DISOC is regulated to the battery charging voltage.



Type vi - Bidirectional rear-buck operation: Here, the relays R_2 and R_A are enabled, and the remaining relays are in the off state. The power available in the motor is extracted, and fed back into the battery. The power flow from the motor into the battery is controlled through the duty ratio of the switch S_{B1} .

IV. COMPONENT DESIGN

The inductor and capacitor design of the DISOC plays a major role in converter design. The inductor value is calculated based on the amount of current ripple content Δi_L present in the inductor current. Similarly, the capacitor value is calculated based on the amount of voltage ripple content, Δv present in the output voltage. The components of the DISOC are designed by considering the C-boost mode of operation. Because the DISOC achieves maximum voltage during this mode of operation

Inductors selection: During on state of the switch, the value of current ripple of the inductor over a cycle is expressed as,

$$\Delta i = \frac{V_2}{L} d_1 T_s + \frac{(V_2 + V_1)}{L} d_2 T_s + \frac{V_1}{L} (d_3 - (d_1 + d_2))$$

From the above equation, the inductor L is selected as,

$$L \geq \frac{V_2 d_1 + (V_1 + V_2) d_2 + V_1 (d_3 - (d_1 + d_2))}{\Delta i f_s}$$

The inductor is chosen by assuming the current ripple of $<10\%$ in CCM operation.

Capacitors selection: The output voltage ripple is lesser than 1% of output voltage V_0 , where equivalent series resistance (ESR) of the capacitor is neglected. The capacitor C is charged during on state of the switch, and the stored energy in the capacitor is written as,

$$Q_C = C \Delta V_C = I_0 (1 - d_3) T_{on}$$

After simplifying, the capacitance is expressed as

$$C \geq \frac{I_0 (1 - d_3)}{f_s \Delta V_0}$$

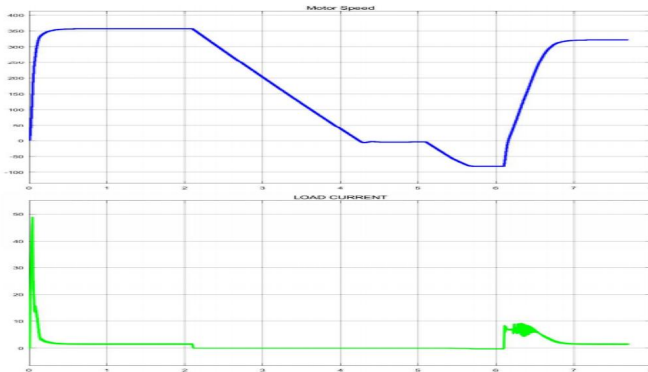
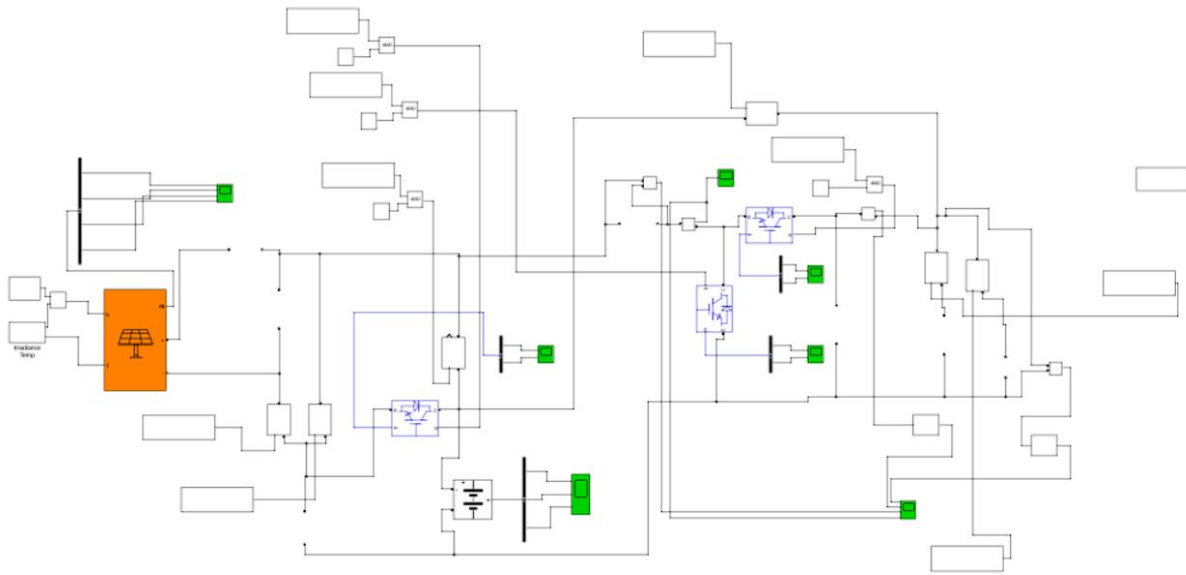
TABLE I SPECIFICATION OF DIFFERENT PARAMETERS OF DISOC

Parameters	Specification			
Source (V_2)	48 V			
Source (V_1)	36 V			
Output voltage (V_0)	156 V			
Capacitor	100 μF			
Inductor	5 mH			
Switching Frequency	20 kHz			
Duty ratios used in different types of operation	$d_1\%$	$d_2\%$	$d_3\%$	$d_{SB1}\%$
i. C-boost operation	20	20	69	off
ii. Solar PV-powered operation	off	off	77	off
iii. Battery-powered operation	off	off	69	off
iv. Battery charging from solar PV	off	off	25	off
v. V2G in parking mode	off	off	77	off
vi. Bidirectional rear-buck operation	off	off	off	54

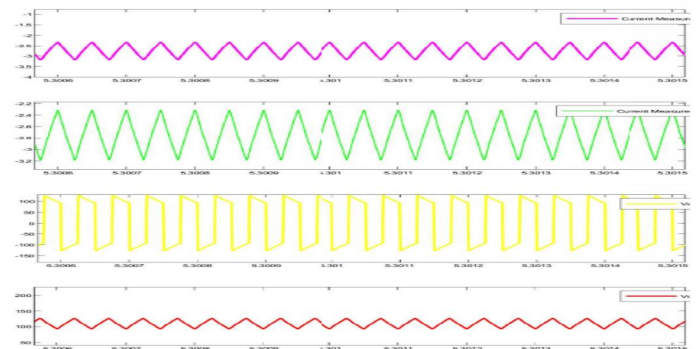
V. RESULTS AND DISCUSSION

The primary source is considered to be a photovoltaic (PV) panel of 10 W (two Mono crystalline 5W PV panel) and a 12 V, 1.3Ah Lead Acid battery is considered to be a storage element. The PV panel is connected with ARDUINO UNO CP2102 and ATMEGA 16A controller units. The controlled switches in the power circuit are realized using MOSFET (IRFP250N). Switch and state selection control algorithms are embedded in the controller. This is simulated in the MATLAB and obtained the results.

Simulink Model :



Motor Characteristics



Regeneration

their body diodes act as boost converter diodes. The motor speed and input battery voltage are used to calculate the duty cycles of the low-side switches of the motor driver

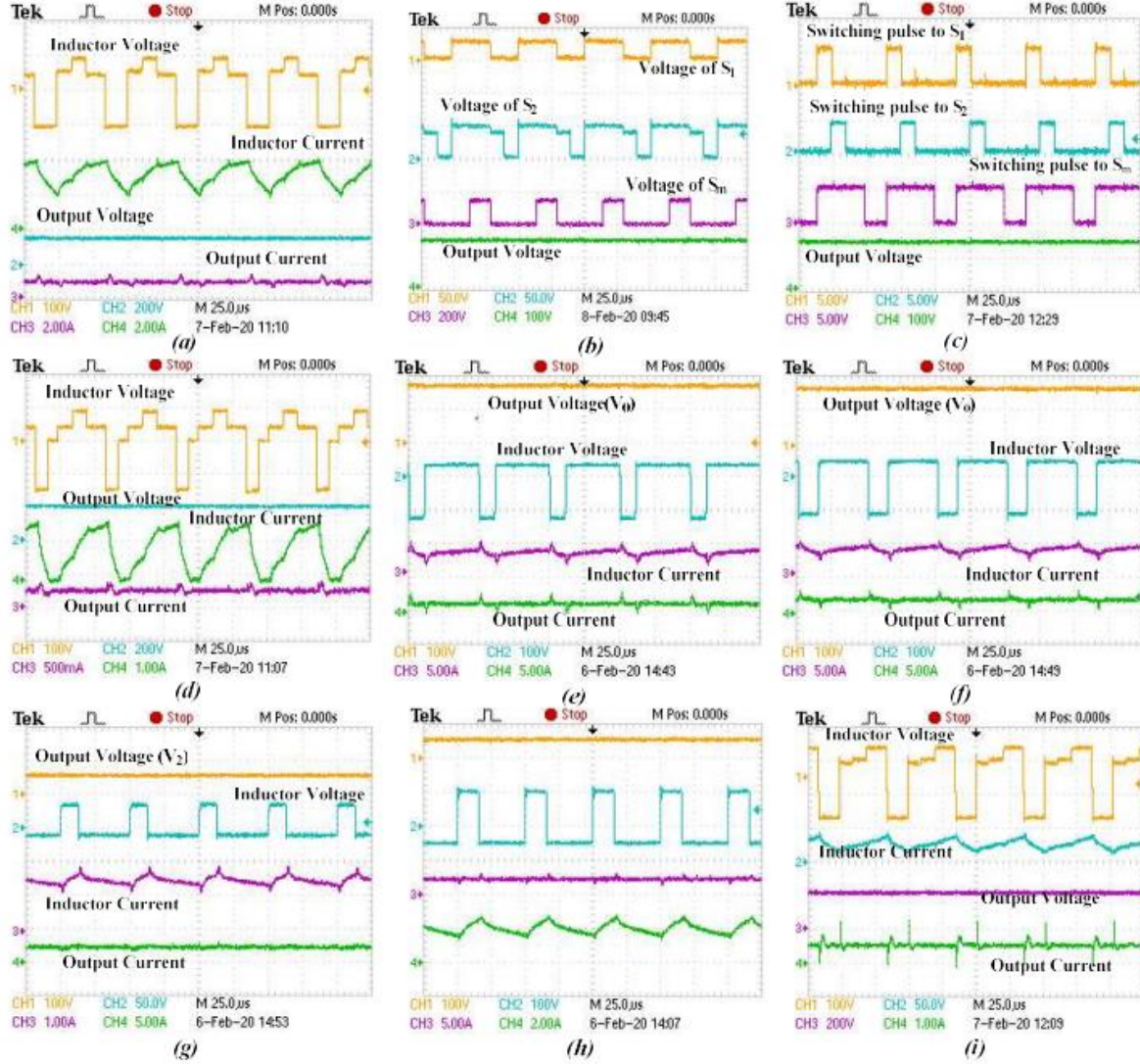


Fig. 12 Experimental results (a) *C-boost* operation in CCM (b) voltage stress in *C-boost* (c) switching pulses in *C-boost* (d) *C-boost* operation in DCM (e) solar PV-powered (f) battery-powered (g) battery charging from solar PV (h) rear-buck (i) *C-boost* when inductor voltage when V_1+V_2 freewheels in mode

V. CONCLUSION

In this project, a new method, called two-boost method, is proposed for the energy regeneration in EVs driven by a BLDC motor. This method is based on converting the BLDC motor driver circuit into two boost converters. This method is based on a similar method, called single boost method, which converts the BLDC motor driver into only one boost converter. In the single-boost method, two phases are used in every 60 electrical degrees while in the two boost method, all the three phases are used at the same time. In both methods, high-side switches are opened, and

in both methods. Implementing the two-boost method requires an additional timer to determine the phase position indices. Based on the simulation and experimental results, the amount of regenerated energy given to the battery in the two-boost method is more than twice that in the single-boost method. Based on the simulation and experimental results, using the two-boost method for the regenerative braking will increase the regenerated energy and system efficiency.

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